

Spin Chains

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1 Preliminaries

1.1 Quantum Mechanical Systems

There are two things that any reasonable physical theory must be able to prescribe to a given system: a state space, and a dynamical law governing how states evolve through time.

In classical physics, the state of any system can be represented mathematically as an element of some real “phase space”; usually the space of solutions to a particular equation of motion. Observable quantities are functionals on this space. There is also a distinguished observable, which we call the Hamiltonian, which governs the evolution of states through time.

In quantum mechanics, the situation is a bit different, and we model systems in this realm according to the following formal axioms:

Axiom. *The state of a quantum mechanical system is given by a nonzero vector in a complex vector space $\mathcal{H}$ endowed with a Hermitian inner product. A general element of this state space will be denoted by $|\psi\rangle$.*

Axiom. *The observables of a quantum mechanical system are given by self-adjoint linear operators on its state space $\mathcal{H}$. There is a distinguished observable, the Hamiltonian $\hat{H}$. Time evolution of states $|\psi(t)\rangle \in \mathcal{H}$ is governed by the Schrödinger equation:*

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle.$$

Axiom. *When an observable A is measured for a quantum system in state $|\psi\rangle$, the result will be an eigenvalue λ_i of A , with probability $\langle \psi | P_{\lambda_i} | \psi \rangle$, where P_{λ_i} is the projection operator onto the eigenspace corresponding to λ_i . Upon measuring the value λ_i , the system will collapse to the state $P_{\lambda_i} |\psi\rangle$.*

So, in contrast with classical systems, the value of an observable in a quantum system is probabilistic, and thus does not have a single well-defined value for each state in the Hilbert space. Note, moreover, that since observables are *self-adjoint* operators, their eigenvalues (i.e., the result of measurements) will always be real numbers, even though the Hilbert space is a complex vector space.

1.2 Group Representations

The symmetries of the real world on a quantum system manifest as actions of certain symmetry groups on its associated Hilbert space. This is expressed mathematically by the language of representations.

Definition 1.2.1. *Given a group G and a vector space V , a **representation of G on V** is a group homomorphism $\pi: G \rightarrow GL(V)$. A representation is called **unitary** if V is also endowed with an inner product $\langle \cdot, \cdot \rangle$ and $\pi(g)$ preserves this inner product for all $g \in G$.*

Remark. *Since a representation requires the datum of a vector space, and it is cumbersome to always write $\pi: G \rightarrow GL(V)$, we shall use the convenient shorthand (π, V) unless the vector space is clear from context.*

Definition 1.2.2. *Let (π, V) be a representation of a group G . If W is a proper, nonzero subspace of V which is invariant under $\pi(g)$ for all $g \in G$, then the restriction of π to W (defined in the obvious way) yields a representation of G on W , and we say this is a **subrepresentation** of π . A representation is said to be **irreducible** if it has no subrepresentations.*

We can also use two representations of a group G to generate a new representation on a “larger” domain, as follows:

Definition 1.2.3. *Given representations (π_1, V_1) and (π_2, V_2) of G , there is a representation $(\pi_1 \oplus \pi_2, V_1 \oplus V_2)$ of G , defined by*

$$(\pi_1 \oplus \pi_2)(g)(v_1, v_2) = (\pi_1(g)(v_1), \pi_2(g)(v_2))$$

It is straightforward to verify that the definition of $\pi_1 \oplus \pi_2$ given above indeed defines a representation of G on $V_1 \oplus V_2$.

Theorem 1.2.1. *Suppose that $\pi: G \rightarrow GL(V)$ is a unitary representation. Then π can be decomposed into a direct sum $\pi = \bigoplus_{j=1}^n \pi_j$ where the π_j are irreducible.*

The above follows from the fact that because π is unitary, the orthogonal complement $W^\perp$ of any $W \subset V$ will be invariant under the action of $\pi(g)$ for all $g \in G$, so any subrepresentation $(\pi|_W, W)$ of V will satisfy $\pi = \pi|_W \oplus \pi|_{W^\perp}$. This process of decomposition can be repeated until we reach a direct sum of irreducibles. The following lemma shows the importance of irreducible representations.

Theorem 1.2.2 (Schur's Lemma). *Suppose that a representation (π, V) of a group G on a complex vector space V is irreducible. Then the only operators on V which commute with every $\pi(g)$ are scalar multiples of the identity.*

Corollary 1.2.1. *If G is an abelian group, all of its irreducible representations are 1-dimensional.*

1.3 Lie groups and their associated Lie algebras

Definition 1.3.1. A **Lie group** is a group that is also a smooth manifold.

Definition 1.3.2. A **Lie algebra** is a vector space V endowed with a Lie bracket $[\cdot, \cdot]$; a bilinear map $V \times V \rightarrow V$ which satisfies the Jacobi identity:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

When we have a Lie group, a natural corresponding Lie algebra arises: the tangent space at the identity. Of course, it's not immediately clear that this is so, for while we know from elementary differential geometry that the tangent space at the identity is a vector space, we have not specified what the Lie bracket should be. However, in this project, we shall only ever be interested in Lie groups which happen to be *matrix* groups, in which case the Lie bracket operation is simply the matrix commutator:

Proposition 1.3.1. *Given a matrix Lie group G with identity E , the set $\mathfrak{g} := T_E G$ together with the Lie bracket $[\cdot, \cdot]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ given by the matrix commutator,*

$$[X, Y] = XY - YX,$$

defines a Lie algebra, which we call the Lie algebra of G .

Remark. *Mathematicians have quite a few ways of thinking about the “tangent space” at a given point of a manifold. The definition we’re using here is the space of all velocity vectors of curves passing through the point of interest; that is, given a manifold M and a point $p \in M$:*

$$T_p M := \{\gamma'(0) \mid \gamma: (-a, a) \rightarrow M \text{ is smooth and satisfies } \gamma(0) = p\}.$$

In the case of a matrix group, we can limit ourselves to only consider curves of the form $\gamma: t \mapsto e^{tY}$, and the tangent space is the space of matrices Y satisfying

$$e^{tY} \in G$$

for all $t \in \mathbb{R}$.

If we have a homomorphism $\pi: G_1 \rightarrow G_2$ of (matrix) Lie groups G_1 and G_2 , there are two important facts about π that are worth noting. The first is that π is (by definition) differentiable. The second is that, given $X \in \mathfrak{g}_1$, the map sending t to $\pi(e^{tX})$ defines a path in G_2 that passes through the identity E_2 of G_2 at $t = 0$. Thus, by definition:

$$\left. \frac{d}{dt} \right|_{t=0} \pi(e^{tX}) \in T_{E_2} G_2 = \mathfrak{g}_2.$$

Since this holds for all $X \in \mathfrak{g}_1$, we get a natural map $\pi': \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$, defined as the differential at the identity:

$$\pi'(X) = \left. \frac{d}{dt} \right|_{t=0} \pi(e^{tX}).$$

Proposition 1.3.2. *π' is a Lie algebra homomorphism; in other words, it is linear and satisfies*

$$\pi'([X, Y]) = [\pi'(X), \pi'(Y)].$$

for all $X, Y \in \mathfrak{g}_1$.

Proposition 1.3.3. A map $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ which is linear and preserves the Lie bracket is called a **Lie algebra representation** of $\mathfrak{g}$ on V . If, in addition, the map ρ satisfies

$$\rho(X)^\dagger = -\rho(X),$$

the representation is said to be **unitary**.

Corollary 1.3.1. A representation $\pi: G \rightarrow GL(m, \mathbb{C})$ of a Lie group G determines a representation π' of its Lie algebra $\mathfrak{g}$, given by

$$\pi'(X) := \left. \frac{d}{dt} \right|_{t=0} \pi(e^{tX}).$$

Proposition 1.3.4. The map π' defined above satisfies the following properties:

1. For all $X \in \mathfrak{g}_1$ and $t \in \mathbb{R}$:

$$\pi(e^{tX}) = e^{t\pi'(X)}.$$

2. For all $X \in \mathfrak{g}_1$ and $g \in G_1$:

$$\pi'(gXg^{-1}) = \pi(g)\pi'(X)\pi(g)^{-1}.$$

The above tells us that we can learn about a Lie group representation $\pi: G \rightarrow GL(V)$ by studying its corresponding Lie algebra representation $\pi': \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. As $\mathfrak{g}$ is a vector space and π' is a linear map, these are much easier to study than Lie group representations. The correspondence becomes even stronger when G is simply connected:

Theorem 1.3.1. If a Lie group G is simply connected, then every Lie algebra representation of $\mathfrak{g}$ integrates to a unique Lie group representation of G .

In other words, if G is simply connected, studying the representations of G is equivalent to studying the representations of its Lie algebra.

1.4 Simplicial complexes

Definition 1.4.1. Let V be a vector space. A subset $C \subset V$ of V is said to be **convex** if for any two points $p, q \in C$, the line segment connecting p and q lies entirely in C ; that is, for any $t \in [0, 1]$ we have

$$tp + (1 - t)q \in C.$$

Definition 1.4.2. Given a vector space V , a finite collection $\{v_0, \dots, v_k\}$ of vectors in V is said to be **convex independent** if the set $\{v_1 - v_0, \dots, v_k - v_0\}$ is linearly independent. Note that this definition does not depend on which vector is labelled v_0 .

Theorem 1.4.1. Let V be a vector space over $\mathbb{R}$, and let $\{v_0, \dots, v_k\}$ be a convex independent set of vectors in V . Then the convex set C generated by $\{v_0, \dots, v_k\}$ is exactly the set of all vectors u of the form $u = \sum_{i=0}^k a_i v_i$ where $a_i \geq 0$ for all i and $\sum_{i=0}^k a_i = 1$. Moreover, every $u \in C$ is uniquely expressible in this form.

Definition 1.4.3. Let V be a vector space over $\mathbb{R}$. The convex set generated by a set $\{v_0, \dots, v_k\}$ of $k + 1$ convex-independent vectors in V is called a (closed) **k -simplex** (or simplex of **dimension k**), and is denoted by $[v_0, \dots, v_k]$. Given a closed k -simplex $[v_0, \dots, v_k]$, its interior, or the set

$$\{v \in [v_0, \dots, v_k] : a_i(v) > 0 \text{ for each } i = 0, \dots, k\}$$

is said to be an **open k -simplex**, and is denoted $(v_0, \dots, v_k)$. If $v \in [v_0, \dots, v_k]$ (or $(v_0, \dots, v_k)$), then the tuple $(a_0, \dots, a_k)$ of numbers with $a_i \geq 0$ and $\sum_{i=0}^k a_i = 1$ such that

$$v = \sum_{i=0}^k a_i v_i$$

are called the **barycentric coordinates** of v . Given a non-empty subset $\{j_0, \dots, j_n\}$ of $\{0, \dots, k\}$, the simplices $[v_{j_0}, \dots, v_{j_n}]$ and $(v_{j_0}, \dots, v_{j_n})$ are called closed and open n -faces of $[v_0, \dots, v_k]$, respectively.

Definition 1.4.4. A **simplicial complex** K is a finite collection of open simplices such that:

1. If $(s) \in K$, then every open face of $[s]$ is in K ;
2. If $(s_1), (s_2) \in K$ and $(s_1) \cap (s_2) \neq \emptyset$, then $(s_1) = (s_2)$.

The **dimension** of K is the maximum of the dimensions of the simplices therein.

Definition 1.4.5. Let K and L be simplicial complexes. A map $\phi: [K] \rightarrow [L]$ is a **simplicial map** if:

1. For each vertex v of K , $\phi(v)$ is a vertex of L ;
2. For each simplex $(v_0, v_1, \dots, v_k) \in K$, the vertices $\phi(v_0), \phi(v_1), \dots, \phi(v_k)$ all lie in some closed simplex in L ; and,
3. ϕ is linear when restricted to each simplex in K .

In other words, a simplicial map preserves the structure of the original complex, but may collapse some simplices into one of a lower dimension. We can also abstract away from the ambient vector space, and consider an *abstract simplicial complex*, in which we begin with a set of vertices $\{v_0, \dots, v_k\}$ (not the elements of a pre-determined vector space) and allow formal linear combinations of these vertices with weights adding up to 1 as possible “points” of the ambient space. This is a purely combinatorial description of the notion of a simplicial complex which still encodes the relative structure of the complex.

2 $SU(2)$ and its (finite-dimensional) representations

2.1 The groups $U(1)$ and $SU(2)$

Proposition 2.1.1. *The set $U(1)$ of all points on the unit circle (in $\mathbb{C}$), together with the operation of complex multiplication, form an abelian group. There are several equivalent formulations of this group. The most common is*

$$U(1) := \{e^{i\theta} : \theta \in [0, 2\pi)\}.$$

Since $U(1)$ is identical to S^1 , it is a Lie group; and perhaps the most simple example thereof.

Proposition 2.1.2. *The set of all 2×2 complex unitary matrices with determinant one, together with the operation of matrix multiplication, form a group. This group is denoted by $SU(2)$ and is called the **special unitary group** of degree 2.*

As with any group, it would be useful to have a more explicit description of the elements of $SU(2)$. To do this, we start with an arbitrary matrix and use the properties specified in the definition above; after some simplifications, we see that

$$SU(2) := \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \in M_{2 \times 2}(\mathbb{C}) : |\alpha|^2 + |\beta|^2 = 1 \right\}.$$

Geometrically, we can identify elements of $SU(2)$ with a pair of complex numbers (α, β) , which we can then identify with points in $\mathbb{R}^4$. Specifically, by the unit determinant condition, elements of $SU(2)$ correspond to elements $(a, b, c, d) \in \mathbb{R}^4$ with $a^2 + b^2 + c^2 + d^2 = 1$, giving us that geometrically, $SU(2)$ can be visualized as S^3 . In particular, $SU(2)$ is also a Lie group.

2.2 Classifying the irreducible representations of $U(1)$

Since $U(1)$ is an abelian group, we know that all of its irreducible representations are 1-dimensional. Thus, any such representation will be a map $\pi: U(1) \rightarrow GL(1, \mathbb{C}) \cong \mathbb{C}^*$. Differentiating, we get

$$\frac{d}{d\theta} \pi(e^{i\theta}) = \lim_{\Delta\theta \rightarrow 0} \frac{\pi(e^{i(\theta+\Delta\theta)}) - \pi(e^{i\theta})}{\Delta\theta} = \pi(e^{i\theta}) \lim_{\Delta\theta \rightarrow 0} \frac{\pi(e^{i\Delta\theta}) - 1}{\Delta\theta} = C\pi(e^{i\theta}), \quad C = \left. \frac{d}{d\theta} \right|_{\theta=0} \pi(e^{i\theta}).$$

It follows from this that $\pi(e^{i\theta}) = e^{iC\theta}$. Then, by the homomorphism property we have

$$e^{iC\theta} = e^{iC(\theta+2\pi)} = e^{iC\theta} e^{iC2\pi} \implies e^{iC2\pi} = 1,$$

so we see that C must in fact be an integer. Thus, we have the following:

Theorem 2.2.1 (Classification of Irreducible Representations of $U(1)$). *For each $k \in \mathbb{Z}$, there is an irreducible representation $\pi_k: U(1) \rightarrow GL(1, \mathbb{C})$ of $U(1)$ given by $\pi_k(e^{i\theta}) = e^{ik\theta}$. Moreover, if $\rho: U(1) \rightarrow GL(1, \mathbb{C})$ is any finite-dimensional irreducible representation of $U(1)$, then $\rho = \pi_k$ for some k .*

We are almost ready to state the main classification result of this section, but first we'll need the following fact about $U(1)$:

Proposition 2.2.1. *Let $\pi: U(1) \rightarrow GL(V)$ be a finite-dimensional representation of $U(1)$. Then, there exists some inner product on V for which π is unitary.*

This follows from the fact that $U(1)$ is compact as the unit circle in $\mathbb{C}$, so we can start with an arbitrary inner product and take an “average” over $U(1)$ by integrating to get an inner product that is invariant under the $U(1)$ action.

Corollary 2.2.1. *If $\pi: U(1) \rightarrow GL(n, \mathbb{C})$ is any finite-dimensional representation of $U(1)$, then there is a basis of $\mathbb{C}^n$ with respect to which $\pi(e^{i\theta})$ has the form*

$$\pi(e^{i\theta}) = \begin{pmatrix} e^{ik_1\theta} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & e^{ik_n\theta} \end{pmatrix}$$

for integers $k_1, \dots, k_n$.

2.3 The Lie algebra $\mathfrak{su}(2)$

Since $SU(2)$ is simply connected, we will instead classify the irreducible representations of its Lie algebra $\mathfrak{su}(2)$. Thus, we begin by giving a more explicit description of the elements of $\mathfrak{su}(2)$, just as we did for $SU(2)$. From the definition of $SU(2)$ and some computations with matrix exponents, we see that $\mathfrak{su}(2)$ is the space of all 2×2 skew-adjoint, traceless complex matrices:

$$\mathfrak{su}(2) = \{X \in M_{2 \times 2}(\mathbb{C}) : X^\dagger = -X \text{ and } \operatorname{tr} X = 0\}.$$

Note that although the elements are matrices with complex coefficients, $\mathfrak{su}(2)$ is still a *real* vector space. As such, another space that will prove to be important is the *complexification* of $\mathfrak{su}(2)$. Normally, the complexification $V_{\mathbb{C}}$ of a vector space V is defined as the Cartesian product $V \times V$, with a general element (u, v) being labeled $u + iv$, together with the usual complex operations:

$$\begin{aligned} (u_1 + iv_1) + (u_2 + iv_2) &:= (u_1 + u_2) + i(v_1 + v_2) \\ (a + ib)(u + iv) &:= (au - bv) + i(av + bu). \end{aligned}$$

In the special case where V is a matrix space we have that $V \cap iV = \{0\}$, and so we can simply identify $(A, B) \in V_{\mathbb{C}}$ with $A + iB$, taken in the usual matrix sum, instead of a formal sum. Here, since $i\mathfrak{su}(2)$ is the space of 2×2 self-adjoint, traceless complex matrices, we also have $\mathfrak{su}(2) \cap i\mathfrak{su}(2) = \{0\}$, so $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{su}(2) + i\mathfrak{su}(2) = \mathfrak{sl}(2; \mathbb{C})$

2.4 Classifying the irreducible representations of $SU(2)$

Suppose we have a representation (π, V) of $SU(2)$. We can consider the corresponding Lie algebra representation of $\mathfrak{su}(2)_{\mathbb{C}}$, and since $SU(2)$ is simply connected, studying the latter is equivalent to studying the former. Here, we pass to the complexification of $\mathfrak{su}(2)$ since it allows us to make use of a simpler basis, namely:

$$S^z := \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S^+ := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S^- := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

To characterize the representations of $SU(2)$, we begin by considering the subgroup thereof traced out by matrices of the form

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

These matrices are unitary and have determinant 1, so they are element of $SU(2)$. Moreover, each such matrix can clearly be identified with the element $e^{i\theta}$ of $U(1)$, and so we have that the collection of these matrices trace out a $U(1)$ subgroup of $SU(2)$. We can now consider the induced $U(1)$ representation given by restricting π to this $U(1)$ subgroup. By the representation theory of $U(1)$, we then have that

$$(\pi_{U(1)}, V) = \mathbb{C}_{q_1} \oplus \cdots \oplus \mathbb{C}_{q_m}$$

for some integers $q_1, \dots, q_m$, where $\mathbb{C}_{q_j}$ denotes the 1-dimensional irreducible $U(1)$ representation corresponding to the integer q_j . One important property of these q_j is that if q is in the set $\{q_j\}$, then so is $-q$. This can be verified by conjugating by a change of basis matrix that permutes the diagonal elements.

Now, choosing a basis of V with respect to this direct sum decomposition, we have that

$$\pi(e^{i2\theta S^z}) = \pi \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} = \begin{pmatrix} e^{ik_1\theta} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & e^{ik_n\theta} \end{pmatrix},$$

from which it follows that

$$\pi'(2iS^z) = \begin{pmatrix} ik_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & ik_n \end{pmatrix}.$$

Dividing both sides of the above equality by $2i$, we then see that

$$\pi'(S^z) = \begin{pmatrix} k_1/2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & k_n/2 \end{pmatrix}.$$

Definition 2.4.1. The integers $k_1, \dots, k_n$ above are called the **weights** of the representation π . The corresponding eigenspace

$$V_k := \{v \in V : \pi'(S^z)v = \frac{k}{2}v\}$$

is referred to as the k th **weight space** of the representation and its dimension $\dim V_k$ is said to be the **multiplicity** of the weight k in the representation.

Using the commutation relations

$$[S^z, S^+] = S^+, \quad [S^z, S^-] = S^-, \quad [S^+, S^-] = 2S^z$$

we obtain that for any $v \in V_k$,

$$\pi'(S^z)\pi'(S^+)v = \pi'(S^z)\pi'(S^+)v + \pi'(S^+)v = (\frac{k}{2} + 1)\pi'(S^+)v$$

and so $\pi'(S^+)v \in V_{k+2}$. Similarly, we have that $\pi'(S^-)v \in V_{k-2}$.

Since S^+ and S^- raise and lower the eigenvalue (with respect to S^z) associated to v , they are called *raising* and *lowering* operators respectively. Note, moreover, that since V is finite-dimensional, there must exist a vector $v \in V_n$ for some n with the property that $\pi'(S^+)v = 0$, and we call such a v a highest weight vector. Now, if the representation is irreducible, we can repeatedly apply $\pi'(S^-)$ to this highest weight vector to get a basis for V , ending at weight $-n$. This classifies all of the finite irreducible representations of $\mathfrak{su}(2)_{\mathbb{C}}$, giving us the following theorem:

Theorem 2.4.1. All finite-dimensional irreducible representations of $SU(2)$ have weights of the form

$$-n, -n+2, \dots, n-2, n$$

for a non-negative integer n .

So, each irreducible representation is characterized by a non-negative integer n , or equivalently a non-negative half integer $n/2$, which we will call the *spin* of the representation. However, we have not yet shown that there exists a corresponding representation for each non-negative integer, and thus we give a construction below using homogeneous polynomials.

For each non-negative integer n , let V_n denote the vector space of **homogeneous polynomials of degree n** in two complex variables; that is, the space of all functions $f: \mathbb{C}^2 \rightarrow \mathbb{C}$ of the form

$$f(z_1, z_2) = a_0 z_1^n + a_1 z_1^{n-1} z_2 + \dots + a_{n-1} z_1 z_2^{n-1} + a_n z_2^n.$$

Given a group G acting on a space M , there is a natural representation of G on the vector space of *functions* on M given via

$$(\pi(g)f)(x) := f(g^{-1} \cdot x).$$

(the g^{-1} appearing above, as opposed to g , is necessary to ensure that π be a homomorphism)

We can restrict this representation to the homogeneous polynomials V_n , noting that V_n is invariant under $\pi(g)$ due to the homogeneity property and the fact that elements of $SU(2)$ are linear operators. Then, some calculations involving the raising and lowering operators show that the resulting representation is an irreducible representation of $SU(2)$ with weights $-n, -n+2, \dots, n-2, n$.

The final step is to give an inner product for which this representation is unitary. Although we know that such an inner product must exist by the same averaging process as before (since $SU(2)$ is compact), we can also give an explicit inner product:

$$\langle f, g \rangle = \frac{1}{\pi^2} \int_{\mathbb{C}^2} \overline{f(z_1, z_2)} g(z_1, z_2) e^{-(|z_1|^2 + |z_2|^2)} dz_1 dz_2.$$

3 Spin Chains

3.1 Introduction

In the quantum world, particles have an intrinsic property called spin, and in the idealized situation where we ignore their spatial motion, we can model the quantum system of the particle with a finite-dimensional irreducible representation of $SU(2)$. We would like to create a mathematical model for a system where there are multiple particles and their spins interact.

By conventional methods, solving such a multi-particle system is a daunting task; since the Hilbert spaces of each particle are tensored together, the dimension of the full Hilbert space of the system grows exponentially as particles are added. Even with the simplest nontrivial case of a spin- $\frac{1}{2}$ particle, the solution to the system becomes impossible to compute by brute force after just 14 particles.

In a spin chain, the particles form a closed loop, where only adjacent particles interact with each other, which gives us translational and reflectional symmetry. Furthermore, the kind of interactions that can occur between particles is limited to one that maintains some sort of symmetry. These symmetries drastically improve the solvability of the quantum system.

Although spin chains with particles described by various representations of different groups are possible, we will limit ourselves to the case where all of the particles are spin- $\frac{1}{2}$. Also, we will only consider the case where the nearest neighbour interaction is identical throughout the spin chain.

3.2 Spin-spin interaction

We first consider a system of two spin- $\frac{1}{2}$ particles interacting with each other. Here, the Hilbert space is $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$, and the interaction between them is prescribed by a Hamiltonian H on this space. For symmetry, we then demand that the Hamiltonian maintains full $\mathfrak{su}(2)$ symmetry with respect to the representation. For this case, this is equivalent to commuting with the spin operator at each site extended to the full space, which are:

$$S^j \otimes I, I \otimes S^j$$

where S^x, S^y, S^z are the spin operators..

Since $(\frac{1}{2}) \otimes (\frac{1}{2}) = (1) \oplus (0)$, the space decomposes into a triplet and a singlet subspace, and by Schur's lemma, H is equal to a scalar product of the identity on all irreducible subrepresentations of the space, giving the form

$$H = aP_{\text{triplet}} + bP_{\text{singlet}}$$

for some real numbers a and b , where P denote projectors into the respective subspaces. Then, noting that $P_{\text{triplet}} + P_{\text{singlet}} = I$ and $\vec{S}_1 \cdot \vec{S}_2 = \frac{1}{4}P_{\text{triplet}} - \frac{3}{4}P_{\text{singlet}}$, we have that the general spin-spin Hamiltonian can also be written as

$$H = -J\vec{S}_1 \cdot \vec{S}_2$$

for a real number J , up to a trivial translation of the Hamiltonian by a scalar multiple of the identity.

3.3 A mathematical model for spin chains

Although the structure of the spin chain is fully determined by the number of particles, as a closed loop, we would like to define spin chains using mathematical objects that better encapsulate the chain structure. To do this, we turn to abstract simplicial complexes, and begin by defining the notion of a “loop 1-complex.”

Definition 3.3.1. A *loop 1-complex* K is an abstract 1-complex of the form

$$K = \{(v_i), (v_i, v_{i+1}) \mid i = 1, \dots, N\}$$

for some positive integer N and vertices $v_1, \dots, v_N$ (we identify $N+1$ with 1 in the indices). The integer N is called the **length** of the loop 1-complex.

Now, we define a spin chain.

Definition 3.3.2. A *spin chain* S is a loop 1-complex K together with a nonzero real number J which prescribes the nearest neighbour interaction.

Definition 3.3.3. Let $\mathcal{S} = (K, J)$ be a spin chain. The Hilbert space of the spin chain is $(\mathbb{C}^2)^{\otimes N}$ where N is the length of K , and the Hamiltonian is

$$H = -J \sum_{j=1}^N \vec{S}_j \cdot \vec{S}_{j+1}$$

where $\vec{S}_j$ denotes the spin operators at site j , extended to the entire space by tensoring with the identity on the other sites.

Note that although we have restricted our focus on the 1-dimensional spin- $\frac{1}{2}$ chain here, the definitions given can easily be extended to the general case, and even to cases where we have higher dimensional spin complexes.

3.4 Morphisms between spin chains

Next, we would like to form spin chains into a category. To do this, we need to define a notion of *morphisms* between these spin chains, and we consider a local approach to defining a morphism, using tools from simplicial complexes.

Definition 3.4.1. Let $\mathcal{S} = (K, J_1)$ and $\mathcal{T} = (L, J_2)$ be spin chains, and let N be the length of $\mathcal{S}$. Then, a spin chain morphism $f: \mathcal{S} \rightarrow \mathcal{T}$ is a simplicial map $f_\Delta: K \rightarrow L$ together with linear maps $T_i: \mathbb{C}^2 \rightarrow \mathbb{C}^2$ for $i = 1, \dots, N$.

Here, f_Δ describes where each site of $\mathcal{S}$ is mapped to, and as a simplicial map it preserves the chain structure of the spin chains, although multiple sites in $\mathcal{S}$ may be collapsed into a single site in $\mathcal{T}$. Then, the maps T_i describe how each site of $\mathcal{S}$ is mapped into a site in $\mathcal{T}$.

Note that this definition will generalize easily if we decide to extend spin chains into spin complexes, with more complicated simplicial complexes as the underlying structure.

4 Density Matrices and Tracing

4.1 Introduction

Our definition of a spin chain morphism is only a local one, and does not give a global map between the Hilbert spaces of the spin chains. If the simplicial map $f_\Delta : K \rightarrow L$ was an “isomorphism”, where each site of K is mapped to a unique site L , we could simply take the tensor product of the operators $T_1, \dots, T_N$, along with a corresponding shift in the indices. However, when we have multiple sites being mapped into a single site, this simple process breaks down. Fundamentally, we need a way of mapping a state in a tensor product Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$ to a state in $\mathcal{H}_1$, partially collapsing the state. This can be done by using density matrices and tracing.

4.2 Pure and mixed states

We begin with the concept of pure and mixed states. A pure state is a single state a quantum mechanical system can be in. Since multiplying a vector in the Hilbert space gives the same state, a pure state is an element of the projective Hilbert space, $P(\mathcal{H}) = (\mathcal{H} - \{0\})/\mathbb{C}^*$. However, in practice, we will use normalized elements of the Hilbert space as pure states, and leave a phase shift ambiguity, which can be carried through the following computations if one requires a fully rigorous treatment of the states as elements of the projective Hilbert space.

In contrast to pure states, a mixed state is a *statistical* ensemble of pure states. A mixed state is described by a set of pure states $\{|\psi_j\rangle\}$ with corresponding proportions $\{p_j\}$ which sum to 1. This represents a collection of quantum systems, such as a beam of many particles, in which p_j of them are in the state $|\psi_j\rangle$.

The concept of a mixed state is different from superposition, which is a purely quantum phenomenon. Suppose we have a beam of spin- $\frac{1}{2}$ particles in the pure state $\frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$, and also a beam of particles in the mixed state where half are in the state $|\uparrow\rangle$ and half are in the state $|\downarrow\rangle$. In both cases, if we measure the spin in the z direction, after sufficiently many measurements, we will measure $\frac{1}{2}$ and $-\frac{1}{2}$ with equal probability. As such, these states may seem like they are representing the same phenomenon, but this is not the case.

Let $|\psi_1\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$ and $|\psi_2\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle$, and consider measuring the spin in the direction for which $|\psi_1\rangle$ and $|\psi_2\rangle$ are eigenstates, with eigenvalues $\frac{1}{2}$ and $-\frac{1}{2}$ respectively. Here, we see a difference between the pure and mixed state. For the pure state, we will always measure $\frac{1}{2}$, but for the mixed state, we will measure $\frac{1}{2}$ and $-\frac{1}{2}$ with equal probability, as both $|\uparrow\rangle$ and $|\downarrow\rangle$ are a superposition of $|\psi_1\rangle$ and $|\psi_2\rangle$ with coefficients with absolute value $\frac{1}{\sqrt{2}}$.

4.3 Density matrices

Mathematically, the concept of a mixed state is expressed using *density matrices*. The density matrix of a pure state $|\psi\rangle$ is given by $|\psi\rangle\langle\psi|$, while the density matrix of a mixed state is the corresponding linear combination of the pure state density matrices. For a mixed state given by pure states $\{|\psi_j\rangle\}$ with corresponding proportions $\{p_j\}$, the corresponding density matrix would then be

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

Here, the matrix ρ encapsulates the idea of the mixed state being a statistical phenomenon, and is very different from the density matrix corresponding to the superposition pure state $\sum_j p_j |\psi_j\rangle$. These density matrices allow us to express mixed states with a single familiar mathematical object, and greatly facilitates the manipulations of mixed states.

4.4 Tracing

We now return to the problem of collapsing a tensor product Hilbert space into one of its constituent Hilbert spaces. For example, consider the problem of assigning a single state to the first particle for the pure (entangled) state $|\psi\rangle = \frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle \in (\frac{1}{2}) \otimes (\frac{1}{2})$. If we measure the spin in the z direction in the first particle, with the observable $S_z \otimes I$, we will measure $\frac{1}{2}$ and $-\frac{1}{2}$ with equal probability. As such, it may be tempting to assign the pure state $|\phi\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$ to the first particle. However, measuring the spin of the first particle in the direction of $|\phi\rangle$ will show that $|\psi\rangle$ behaves differently than the pure state $|\phi\rangle$.

In fact, measurements on the first particle of $|\phi\rangle$ behaves exactly as if it is in the mixed state $\frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow|$, and this is precisely why mixed states are useful.

Now, consider an arbitrary tensor product Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$, where $\mathcal{H}_1$ and $\mathcal{H}_2$ are Hilbert spaces with dimensions n and m respectively, and we wish to create a map into $\mathcal{H}_1$ that collapses the $\mathcal{H}_2$ component. We also want to maintain as much of the quantum nature in the $\mathcal{H}_1$ part of the original state as possible. To do this, we choose an orthonormal basis $(f_1, \dots, f_m)$, and collapse the state by measuring by an $\mathcal{H}_2$ observable with the f_i 's as eigenvectors, which we denote O .

Let $|\psi\rangle$ be a normalized state in $\mathcal{H}_1 \otimes \mathcal{H}_2$, and let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of O corresponding to $f_1, \dots, f_m$ respectively. Then, we perform a measurement by the observable $I \otimes O$, which will give one of the λ_j 's as the result, with probability given by the Born rule:

$$\begin{aligned} p_j &= \langle \psi | P_{\lambda_j} | \psi \rangle \\ &= \langle \psi | (I \otimes |f_j\rangle\langle f_j|) | \psi \rangle \\ &= \langle \psi | (I \otimes |f_j\rangle\langle f_j|) (I \otimes \langle f_j|) | \psi \rangle \\ &= |(I \otimes \langle f_j|) | \psi \rangle|^2 \end{aligned}$$

where P_{λ_j} denotes the projection to the λ_j eigenspace of $I \otimes O$.

Upon measuring λ_j , the state will collapse to the state $P_{\lambda_j} |\psi\rangle$, which is an element in the eigenspace $\mathcal{H}_1 \otimes \text{span}(|f_j\rangle)$. This space can then be identified with $\mathcal{H}_1$, since $\text{span}(|f_j\rangle)$ is 1-dimensional, with the natural identification done by applying $I \otimes \langle f_j|$. So, combining these maps, we have that $|\psi\rangle$, upon measuring λ_j , collapses to

$$(I \otimes \langle f_j|) P_{\lambda_j} |\psi\rangle = (I \otimes \langle f_j|) (I \otimes |f_j\rangle\langle f_j|) |\psi\rangle = (I \otimes \langle f_j| f_j \rangle \langle f_j|) |\psi\rangle = (I \otimes \langle f_j|) |\psi\rangle$$

Then, we renormalize this state, to get

$$\frac{1}{|(I \otimes \langle f_j|) |\psi\rangle|} (I \otimes \langle f_j|) |\psi\rangle$$

Finally, we do this process for each of the possible measurements results λ_j , and put them together into a density matrix in $\mathcal{H}_1$:

$$\begin{aligned} \rho_{\mathcal{H}_1} &= \sum_{j=1}^m p_j \cdot \frac{1}{|(I \otimes \langle f_j|) |\psi\rangle|} (I \otimes \langle f_j|) |\psi\rangle \cdot \frac{1}{|(I \otimes \langle f_j|) |\psi\rangle|} \langle \psi | (I \otimes |f_j\rangle) \\ &= \sum_{j=1}^m |(I \otimes \langle f_j|) |\psi\rangle|^2 \cdot \frac{1}{|(I \otimes \langle f_j|) |\psi\rangle|^2} (I \otimes \langle f_j|) |\psi\rangle \langle \psi | (I \otimes |f_j\rangle) \\ &= \sum_{j=1}^m (I \otimes \langle f_j|) |\psi\rangle \langle \psi | (I \otimes |f_j\rangle) \end{aligned}$$

Recalling the definition of the trace of an operator, which is

$$\text{tr}(T) = \sum_k \langle g_k | T | g_k \rangle$$

for any orthonormal basis g_j , this form may be familiar. Indeed, we may express the process of collapsing the $\mathcal{H}_2$ component as taking a trace over just the $\mathcal{H}_2$ component, in two different ways.

First, after choosing an orthonormal basis $(f_1, \dots, f_m)$ of $\mathcal{H}_2$, we can view elements of $\mathcal{H}_1 \otimes \mathcal{H}_2$ as vectors in $\mathcal{H}_\infty$ with coefficients in $\mathcal{H}_1$. Then, an operator on $\mathcal{H}_1 \otimes \mathcal{H}_2$ can be viewed as an $(m \times m)$ matrix on $\mathcal{H}_2$ with entries which are operators on $\mathcal{H}_1$. Then, similar to how $\langle g_p | M | g_q \rangle$ selects the matrix entry in row p and column q for normal matrices, the operator valued entry in row p and column q can be selected as follows:

$$(I \otimes \langle f_p|) |\psi\rangle \langle \psi | (I \otimes |f_q\rangle)$$

Then, taking the “trace” of this operator valued matrix, by summing the diagonal entries, we get

$$\sum_{j=1}^m (I \otimes \langle f_j|) |\psi\rangle \langle \psi | (I \otimes |f_j\rangle)$$

which is exactly the expression for $\rho_{\mathcal{H}_1}$ that we previously obtained. So, with this process, we are taking a *partial trace* over $\mathcal{H}_2$. Also, noting that operators and scalars commute, we can show that this partial trace operation is independent of the orthonormal basis $(f_1, \dots, f_m)$ chosen, by the same argument as how basis independence is shown for the usual operator trace. Thus, we have a well defined map

$$\text{tr}_{\mathcal{H}_2}(\rho_{\mathcal{H}_1 \otimes \mathcal{H}_2}) = \sum_{j=1}^m (I \otimes \langle f_j |) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (I \otimes |f_j\rangle)$$

that maps an operator (density matrix) $\rho_{\mathcal{H}_1 \otimes \mathcal{H}_2}$ on $\mathcal{H}_1 \otimes \mathcal{H}_2$ to an operator (density matrix) on $\mathcal{H}_1$.

Another way to see the connection to trace is to consider the resulting density matrix entrywise. Choosing an orthonormal basis $(e_1, \dots, e_n)$ of $\mathcal{H}_1$ to represent the matrix with, we have that the p, q 'th entry is $\langle e_p | \rho_{\mathcal{H}_1} | e_q \rangle$. Then, we have

$$\begin{aligned} \langle e_p | \rho_{\mathcal{H}_1} | e_q \rangle &= \langle e_p | \sum_{j=1}^m (I \otimes \langle f_j |) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (I \otimes |f_j\rangle) | e_q \rangle \\ &= \langle e_p | \left(\sum_{j=1}^m (I \otimes \langle f_j |) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (I \otimes |f_j\rangle) \right) | e_q \rangle \\ &= \sum_{j=1}^m (\langle e_p | \otimes \langle f_j |) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (|e_q\rangle \otimes |f_j\rangle) \\ &= \sum_{j=1}^m \langle f_j | \left((\langle e_p | \otimes I) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (|e_q\rangle \otimes I) \right) | f_j \rangle \\ &= \text{tr} \left((\langle e_p | \otimes I) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (|e_q\rangle \otimes I) \right) \end{aligned}$$

So, we see that the p, q 'th entry in the resulting density matrix is given by the trace of the operator $(\langle e_p | \otimes I) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (|e_q\rangle \otimes I)$. Here, we are viewing operators on $\mathcal{H}_1 \otimes \mathcal{H}_2$ can be viewed as an $(n \times n)$ matrix on $\mathcal{H}_1$ with entries which are operators on $\mathcal{H}_n$, and taking the operator trace entrywise to obtain the resulting density matrix.

Now that we have a well defined partial trace, we can use it to map a *mixed* state in $\mathcal{H}_1 \otimes \mathcal{H}_2$ to a mixed state in $\mathcal{H}_1$, by simply applying $\text{tr}_{\mathcal{H}_2}$ to the density matrix of the mixed state. This indeed gives what we expect, since

$$\begin{aligned} \text{tr}_{\mathcal{H}_2}(\rho_{\mathcal{H}_1 \otimes \mathcal{H}_2}) &= \sum_{j=1}^m (I \otimes \langle f_j |) \rho_{\mathcal{H}_1 \otimes \mathcal{H}_2} (I \otimes |f_j\rangle) \\ &= \sum_{j=1}^m (I \otimes \langle f_j |) \left(\sum_k p_k |\psi_k\rangle \langle \psi_k| \right) (I \otimes |f_j\rangle) \\ &= \sum_k p_k \sum_{j=1}^m (I \otimes \langle f_j |) |\psi_k\rangle \langle \psi_k| (I \otimes |f_j\rangle) \\ &= \sum_k p_k \text{tr}_{\mathcal{H}_2}(|\psi_k\rangle \langle \psi_k|) \end{aligned}$$

which is simply the density matrix given by collapsing each of the constituent pure states in the mixed state ensemble, then taking the linear combination of the results according to their original proportions.

5 Solving Spin Chains

5.1 Introduction

Now, we take a detour from globalization of the spin chain morphism, and consider spin chains as a quantum system. The most important computations that can be made of a spin chain is diagonalizing the Hamiltonian: finding the eigenstates and their corresponding eigenvalues, which are the energy levels. As mentioned previously, diagonalizing by brute force becomes exponentially harder as nodes are added, and is infeasible past 14 nodes. However, by using the symmetries of the spin chain and the Bethe ansatz, spin chains of many many more nodes can be solved, which we show in this section.

5.2 Using the symmetries of the spin chain

Let (K, J) be a spin chain and let N be its length. The first symmetry that we will consider is the total spin in the z -direction, given by

$$S_T^z = \sum_{j=1}^N S_j^z$$

where S_j^z denotes the spin operator in the z direction for the node j , extended to the entire space by tensoring with the identity. Then, since the spin chain Hamiltonian H commutes with the spin operators at each site, we have that H commutes with S_z , and thus they are simultaneously diagonalizable. H is then block diagonal with respect to the eigenspaces of S_z , which are characterized by the number of down spins M , and we can consider each of these subspaces separately.

Also, we can use another symmetry of the spin chain: the spin flip symmetry. This is given by the spin flip operator

$$F = \sum_{\alpha_1, \dots, \alpha_N = +, -} S_1^{\alpha_1} \dots S_N^{\alpha_N}$$

which flips the spin in the z direction at each node. For example, the state $|\uparrow\downarrow\rangle$ would be mapped to $|\downarrow\uparrow\rangle$. This spin flip operator maps states in the $M = m$ subspace to the $M = N - m$ subspace. Furthermore, since H commutes with all of the S_j^x and S_j^y , and we have $S_j^+ = S_j^x + iS_j^y$ and $S_j^- = S_j^x - iS_j^y$, it also commutes with all of the S_j^+ and S_j^- . Thus, H commutes with F , and we have that an eigenstate of H in the $M = m$ subspace is mapped to another eigenstate of H in the $M = N - m$ subspace, with the same eigenvalue. Therefore, it suffices to find the eigenvectors and eigenvalues of H for the subspaces $M = 0, \dots, \frac{N}{2}$.

To make computation of the Hamiltonian easier, we rewrite it in terms of operators that are easy to compute on $|\uparrow\rangle$ and $|\downarrow\rangle$, using the fact that $S_j^+ = S_j^x + iS_j^y$ and $S_j^- = S_j^x - iS_j^y$:

$$\begin{aligned} H &= -J \sum_{j=1}^N \vec{S}_j \cdot \vec{S}_{j+1} \\ &= -J \sum_{j=1}^N (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + S_j^z S_{j+1}^z) \\ &= -J \sum_{j=1}^N \left(\frac{1}{2} (S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+) + S_j^z S_{j+1}^z \right) \end{aligned}$$

The $M = 0$ subspace consists of a single state, the state with all up spins $|\uparrow \dots \uparrow\rangle = |0\rangle$. This is an eigenstate with eigenvalue $E_0 = -\frac{JN}{4}$. Note that for the ferromagnetic spin chain, where we have $J > 0$, this state is a ground state. Also, this state is decomposable, which means that each of the particles is individually in a well defined state. However, for the antiferromagnetic chain, where we have $J < 0$, the ground state is an entangled state, and it is not possible to assign a pure state to individual particles. This inability to “classically” assign a state to each of the particles for the system’s ground state makes the antiferromagnetic chain more “interesting” of an object than the ferromagnetic counterpart.

Next, we consider the $M = 1$ subspace. Defining the S_T^z eigenvectors

$$|n\rangle = S_-^{(n)}|0\rangle$$

we have that these form a basis, and the space is N -dimensional. However, unlike in the $M = 0$, these natural basis vectors are not eigenvectors of H , and H is not yet diagonalized. So, we introduce a new symmetry: translational symmetry. The translation operator T shifts the spins in each of the nodes by one node, meaning $T|n\rangle = |n+1\rangle$ and $T|N\rangle = |1\rangle$. Since both H and S_T^z are defined by taking a sum over all of the nodes, they commute with T , and the three operators are all simultaneously diagonalizable.

We have that $T^N = I$, so the eigenvalues must be in the form e^{ik} where $k = 0, \frac{2\pi}{N}, \dots, \frac{2\pi}{N}(N-1)$. With these eigenvalues in mind, we can see that the eigenvectors must be in the form

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ikn} |n\rangle$$

for $k = 0, \frac{2\pi}{N}, \dots, \frac{2\pi}{N}(N-1)$. Here, we have N eigenvectors of T with distinct eigenvalues, and thus this choice of basis must also diagonalize H , for the $M = 1$ subspace. Computing H applied to these states, we then have that the energy levels are

$$E - E_0 = J(1 - \cos k)$$

5.3 Introduction to the Bethe ansatz

We can apply the same process to the $M = 2$ subspace, to decompose it into eigenspaces of T . However, these eigenspaces will no longer be 1-dimensional (for $N > 3$), and this will not be enough to diagonalize the Hamiltonian. Even introducing the reflectional symmetry of the spin chain, or the full $\mathfrak{su}(2)$ symmetry is not sufficient to diagonalize the Hamiltonian, and we need a different approach, which is the Bethe ansatz. To show the idea of the Bethe ansatz, we revisit the $M = 1$ case.

A general vector in this subspace is given by

$$|\psi\rangle = \sum_{n=1}^N a(n) |n\rangle$$

for some constants $a(n)$, under the periodic boundary conditions $a(n) = a(n+N)$. The eigenvalue equation $H|\psi\rangle = E|\psi\rangle$ then results in the system of equations

$$2(E - E_0) = J(2a(n) - a(n-1) - a(n+1))$$

for $n = 1, \dots, N$. Then, the Bethe ansatz postulates that these coefficients are in the form $a(n) = e^{ikn}$ where $k = 0, \frac{2\pi}{N}, \dots, \frac{2\pi}{N}(N-1)$ which gives us, after normalization, the eigenvectors previously obtained.

To explain the origin of the form of the coefficients given by the ansatz, we consider the translation operator T . Here, T is a group action which is prescribing the a discrete unit of translation of a down spin on this space. Giving this pseudoparticle the name *magnon*, we can look at the spin chain system from the lens of the mechanics of a particle moving in space. Usually, if we have a translational group action on a particle, we can take a logarithm to obtain a self adjoint infinitesimal translational operator, which is the observable corresponding to momentum (with respect to motion on the space by the aforementioned translation). However, due to the discrete geometry of the spin chain structure, T forms a group structure of C_N , as opposed to a continuous group. As such, we are unable to take a logarithm to obtain a self adjoint operator which is the infinitesimal counterpart. Nonetheless, we can still perform a Fourier expansion to obtain what the numerical value for the “momentum” of the magnon, which is the k that can take values in $0, \frac{2\pi}{N}, \dots, \frac{2\pi}{N}(N-1)$. Then, exponentiating these momentum values to return to the spin chain system gives us the form of the ansatz, for $M = 1$.

5.4 Two magnon states

Now, we solve the Hamiltonian for the $M = 2$ case. Similarly to before, we define the S_T^z eigenvectors

$$|n_1, n_2\rangle = S_-^{(n_1)} S_-^{(n_2)} |0\rangle$$

and we have that these vectors, for $1 \leq n_1 < n_2 \leq N$, form a basis, and the space is $\binom{N}{2}$ -dimensional. Then, a general vector in the $M = 2$ subspace is given by

$$|\psi\rangle = \sum_{1 \leq n_1 < n_2 \leq N} a(n_1, n_2) |n_1, n_2\rangle$$

Then, the Bethe ansatz postulates that these coefficients are in the form

$$a(n_1, n_2) = A e^{i(k_1 n_1 + k_2 n_2)} + A' e^{i(k_2 n_1 + k_1 n_2)}$$

This looks similar to the Bethe ansatz for $M = 1$, but with some important differences. We can once again Fourier expand the system with respect to the translation operator T , but this only gives the total momentum of both the magnons. Ideally, this system would be simply just two superimposed magnons from the 1-magnon case, where the state space would be a tensor product of two copies of the 1-magnon state. This would give us an individual translation operator for each copy of the spin chain, which we could Fourier expand by and get the momenta of each magnon

However, identifying these superimposed states back to the original spin chain Hilbert space would result in an overcomplete set of $\binom{n+1}{2}$ states, which are not eigenstates, and some of which violate the requirement that the two down spins must be at different sites, and are not elements of $M = 2$ as a result.

Then, to embed this $M = 2$ subspace in such a tensor product of multiple spin chains requires some more effort. We first identify the $M = 2$ subspace with the second exterior power of the $M = 1$ subspace, $\bigwedge^2(\mathcal{H}_{M=1})$, which we take to be a subspace of $(\mathcal{H}_{M=1})^{\otimes 2}$. This is done by identifying the state $|n_1, n_2\rangle$ for $n_1 < n_2$ with the alternating tensor $|n_1\rangle \otimes |n_2\rangle - |n_2\rangle \otimes |n_1\rangle \in (\mathcal{H}_{M=1})^{\otimes 2}$, and extending linearly. Then, we Fourier expand by the translation operator at each of the sites, to get the individual momenta. Next, we appropriately pull back these momenta to the $M = 2$ subspace, and we reimpose the alternating condition, which leads to the two different terms in the ansatz (corresponding to the two possible permutations of $\{1, 2\}$). Finally, we let the coefficients of the terms be arbitrary, to be able to account for the effects of the two magnons interacting with each other later in the solution process.

With the ansatz for the coefficients in hand, we continue with finding the eigenvectors. The eigenvalue equation $H|\psi\rangle = E|\psi\rangle$ results in the following $\binom{n}{2}$ equations:

$$2(E - E_0)a(n_1, n_2) = J(4a(n_1, n_2) - a(n_1 - 1, n_2) - a(n_1 + 1, n_2) - a(n_1, n_2 - 1) - a(n_1, n_2 + 1))$$

for $n_1 + 1 < n_2$, and

$$2(E - E_0)a(n_1, n_2) = J(2a(n_1, n_2) - a(n_1 - 1, n_2) - a(n_1, n_2 + 1))$$

for $n_1 + 1 = n_2$. These equations are satisfied as long as the energy depends on the k_1 and k_2 by

$$E - E_0 = J \sum_{j=1}^2 (1 - \cos k_j)$$

and the amplitude ratio satisfies

$$\frac{A}{A'} = e^{i\theta_{1,2}} = -\frac{e^{k_1+k_2} + 1 - 2e^{ik_1}}{e^{k_1+k_2} + 1 - 2e^{ik_2}}$$

This ratio requirement can then be included in the Bethe ansatz by

$$a(n_1, n_2) = e^{i(k_1 n_1 + k_2 n_2) + (1/2)\theta_{1,2}} + A' e^{i(k_2 n_1 + k_1 n_2) + (1/2)\theta_{2,1}}$$

where the phase angle $\theta_{1,2} = -\theta_{1,2}$ is determined by k_1 and k_2 by the equation above. Here, this factor represents the phenomenon of the phase shift picked up by the magnons as they interact with each other.

Noting that

$$e^{i\theta} = \frac{\cot \frac{\theta}{2} + i}{\cot \frac{\theta}{2} - i}$$

we can rewrite the equation relating $\theta_{1,2}$, k_1 and k_2 in a more convenient real form, as

$$2 \cot \frac{\theta_{1,2}}{2} = \cot \frac{k_1}{2} - \cot \frac{k_2}{2}$$

Now, imposing the periodic boundary conditions $a(n_1, n_2) = a(n_2, n_1 + N)$ on the ansatz expression, we have that

$$Nk_1 = 2\pi a_1 + \theta_{1,2}, Nk_2 = 2\pi a_2 + \theta_{2,1}$$

where a_1, a_2 are integers taking values in $\{0, 1, \dots, N-1\}$ and are called the Bethe quantum numbers. These two sets of equations are then called the Bethe equations.

Another way to express the Bethe equations is obtained by reparametrizing them in terms of the rapidity $\lambda_j = \frac{1}{2} \cot \frac{k_j}{2}$. Here, the periodic boundary conditions become the equations

$$\begin{aligned} \left(\frac{\lambda_1 + \frac{i}{2}}{\lambda_1 - \frac{i}{2}} \right)^N &= \frac{\lambda_1 - \lambda_2 + i}{\lambda_1 - \lambda_2 - i} \\ \left(\frac{\lambda_2 + \frac{i}{2}}{\lambda_2 - \frac{i}{2}} \right)^N &= \frac{\lambda_2 - \lambda_1 + i}{\lambda_2 - \lambda_1 - i} \end{aligned}$$

Then, the final step to solving the Hamiltonian for the $M = 2$ is to find all (a_1, a_2) pairs that give solutions to the Bethe equations. By symmetry, we can assume that $a_1 \leq a_2$, and we must find $\binom{n}{2}$ solutions with the $\binom{n+1}{2}$ possible pairs. Using the reparametrization above, this problem is then a matter of solving a system of polynomial equations. This demonstrates the power of the Bethe ansatz, as the hugely complex problem of solving a multi-particle system has been reduced to solving a system of polynomial equations.

Calculating the actual solutions, we get three classes of solutions. Class C_1 are the solutions for which $a_1 = 0$, and these behave exactly like the 1-magnon states. Class C_2 are the 2-magnon scattering states, and are the solutions for which $a_1 + 1 < a_2$ so the two magnons are sufficiently “far apart” from each other to behave very close to two superimposed magnons. All possible such pairs for C_1 and C_2 yield distinct solutions with real momenta.

Finally, class C_3 are the 2-magnon bound states, and are the solutions for which $a_1 = a_2$ or $a_1 + 1 = a_2$. These states behave very differently than the class C_1 or C_2 states. Although there are $2N - 3$ such pairs, only $N - 3$ of them yield solutions. Also, the energy levels of the bound states lie in a single branch below the distribution of the 2-magnon scattering states. Furthermore, the effects of the magnon interaction in class C_3 states do not disappear in the limit $N \rightarrow \infty$, while the effect vanishes for class C_2 states.

5.5 The general case

Finally, the Bethe ansatz can be extended to all cases. For the case $M = m$, we begin as before by defining the S_T^z eigenvectors:

$$|n_1, \dots, n_m\rangle = S_-^{(n_1)} \dots S_-^{(n_m)} |0\rangle$$

and a general vector has the form

$$|\psi\rangle = \sum_{1 \leq n_1 < \dots < n_m \leq N} a(n_1, \dots, n_m) |n_1, \dots, n_m\rangle$$

Then, by the same process as in the $M = 2$ case, but with m magnons as opposed to 2, we get the ansatz

$$a(n_1, \dots, n_m) = \sum_{\mathcal{P} \in S_m} \exp \left(i \sum_{j=1}^m k_{\mathcal{P}_j} n_j + \frac{i}{2} \sum_{p < q} \theta_{\mathcal{P}_p \mathcal{P}_q} \right)$$

where the sum $\mathcal{P} \in S_m$ is over all $m!$ permutations of the set $\{1, \dots, m\}$, corresponding to the alternating condition on $\bigwedge^m(\mathcal{H}_{M=1})$.

Now, solving the eigenvalue equation $\mathcal{H}|\psi\rangle = E|\psi\rangle$ gives the energy

$$E - E_0 = J \sum_{j=1}^m (1 - \cos k_j)$$

and the phase shift equations

$$2 \cot \frac{\theta_{p,q}}{2} = \cot \frac{k_p}{2} - \cot \frac{k_q}{2}$$

for all distinct $p, q = 1, \dots, m$.

Finally, the periodic boundary conditions give the equations

$$Nk_p = 2\pi a_p + \sum_{q \neq p} \theta_{qp}$$

for $p = 1, \dots, m$, where $a_p \in \{0, 1, \dots, N-1\}$ are the Bethe quantum numbers. Reparametrizing in terms of the rapidity λ_p once again, we have

$$\left(\frac{\lambda_p + \frac{i}{2}}{\lambda_p - \frac{i}{2}} \right)^N = \prod_{q \neq p} \frac{\lambda_p - \lambda_q + i}{\lambda_p - \lambda_q - i}$$

for $p = 1, \dots, m$. Once again, the problem of solving the spin chain is reduced to solving a system of polynomial equations.